

OLUWASEGUN ADEWOLA

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EDUCATION

Northeastern University

Master of Science in **Computer Science**, GPA: **4.0**

Expected Dec. 2027

Vancouver, Canada

- Relevant Coursework: Distributed Systems, Computer Vision, Machine Learning, Databases, Object-Oriented Programming, Data Structures and Algorithms, Advanced Algorithms
- People's Choice **Winner**, **Qualcomm & Microsoft** Edge AI Hackathon; **Global Learner Scholarship**

SKILLS

Languages: GoLang, Java, Python, C++, C, JavaScript, TypeScript, Bash, SQL, HTML/CSS

Frameworks: Spring/Spring Boot, Django, FastAPI, Node.js, React.js, Next.js

Data Systems: PostgreSQL, MySQL, Redis, Cassandra, MongoDB, DynamoDB, S3, SQS, RabbitMQ, Kafka

AI/ML: LangChain, LangGraph, ChromaDB, PyTorch, scikit-learn, NumPy, Streamlit

Tools: Linux, AWS, GCP, Nginx, Docker, Kubernetes, Git, GitHub Actions, Terraform, Prometheus, Grafana, Gradle, npm

PROJECTS

Chatflow (Distributed Chat App)

[GitHub](#)

- Built a distributed chat system with **4** horizontally-scaled **Java Spring Boot WebSocket** servers on **AWS EC2** behind a load balancer, with **RabbitMQ** for message routing, **Redis** for caching, **Cassandra** for storage, and dedicated consumer processes decoupling message delivery from persistence.
- Achieved **13K messages/second** throughput validated through load and integration testing, with dead letter queues, exponential backoff, and circuit breaker patterns for production-grade fault tolerance.

RemitTrack (Skyscanner for Currency Exchange Rates)

[Architecture](#)

- Built a real-time exchange rate platform for the African diaspora covering **15+** Money Transfer Providers; a **GoLang** modulith on **EC2/NGINX** serves **REST APIs**, price-alert notifications via **AWS SES**, and a **B2B** data endpoint, backed by **RDS PostgreSQL**, **ElastiCache**, **WAF**, and **Stripe**.
- Serverless **GoLang Lambda** functions triggered by 5-minute **EventBridge** crons emulate browser TLS fingerprints using **uTLS** and route through rotating residential proxies to bypass bot detection, pushing normalized rate records to **SQS** for idempotent ingestion into **RDS PostgreSQL**.
- Delivered a **Next.js**, **TypeScript**, and **React** frontend compiled to flat static files, hosted on **S3**, and edge-served via **CloudFront** with **ISR** webhooks refreshing corridor pages on every ingestion cycle at near-zero origin compute for page traffic.
- Engineered a **CI/CD** pipeline in **GitHub Actions** built on **Test-Driven Development**, running **GoLang** unit tests and a **Docker Compose** sandbox before shipping via **Terraform**-provisioned infrastructure and a zero-downtime **blue/green** rolling swap.

Coast (Agentic Self-Healing & Compliance Harness for RemitTrack)

[Architecture](#)

- Built a **multi-agent** self-healing pipeline in **Python**, **LangGraph**, and **ChromaDB**; on scraper failure, an agent retrieves similar past fixes via **RAG** over a **ChromaDB** vector store, has an **LLM** generate a patch, and opens a pull request, cutting recovery time by **97%**.
- Each **AI**-generated patch passes a **GoLang AST** safety linter, runs in a **Docker Compose** sandbox against a regression corpus of past failures, and clears a **3-sigma** statistical canary check before opening a pull request for human review.

EXPERIENCE

Graduate Teaching Assistant

Aug. 2025 – Present

Khoury College of Computer Science, Northeastern University

Vancouver, Canada

- Supported **100+** students in Prof. Coady's **Distributed Systems** course through labs and office hours covering consensus, microservices, scaling, observability, and concurrency, using **Java**, **GoLang**, **Terraform**, **AWS**, **RabbitMQ**, **Docker**, and **Redis**, contributing to zero resubmissions and positive feedback.
- Led instruction for **40+** students in **Discrete Math** (Fall 2025) and **50+** students in **Data Structures and Algorithms** (Spring 2026) in **C**, through teaching, office hours, and grading under Jelani Drew and Prof. Mullen respectively, contributing to a **100%** course pass rate.

Software Engineer

Dec. 2022 – Aug. 2024

Flutterwave YC16 (Digital Payments for African businesses)

Lagos, Nigeria

- Conceived and independently delivered the predictive pricing service for **SendApp**, Flutterwave's consumer FX product, as sole engineer on a cross-functional initiative, ensuring SendApp surfaced the best affordable rate across **500+** currency pairs and driving a **66%** revenue increase in month one.
- Designed dynamic pricing algorithms in **Python** using **Linear Programming**, validated by domain experts, and built **Power BI** dashboards for real-time error detection; independently monitored SendApp's own FX rates for cross-currency arbitrage exposure, together saving **\$1.5M+** in revenue over 6 months.
- Built serverless scrapers using **Playwright** or **Requests** per each competitor's **ToS/robots.txt**, cleaned data at scale in **Python**, and loaded it into **Google Sheets** for Finance, Forex, and Accounting teams via scheduled **GitHub Actions** jobs.